

Exam Information and Preparation Exercises

Examination format:

Oral exam. 6-minute presentation + 9 minutes of Q&A.

Exam aids: Students may use:

- "presenter notes" in their presentations
- "backup slides" they anticipate may be useful during the Q&A
- any notes, slides or material generated during the exam preparation exercises during the course
- the Miro board with collective notes taken during the course: https://miro.com/app/board/uXjvG_0_T-A=?share_link_id=928435085081 (password: mhci26aau)

Note: While students can bring these aids to the exam, please have in mind that we expect you to already know the material. The 15 minutes allocated to the exam are for demonstrating your knowledge and not for reading material for the first time.

Exam procedure

Exam Part 1

You must present a 6-minute presentation including:

1. An overview of a paper of your choice (see "[Exam papers](#)" below), clearly describing:
 - the research problem and motivation
 - the key design idea being studied (only if the paper is proposing a new design, feature, technique, etc.)
 - the user study methods
 - the main findings
2. Your own idea for applying the takeaways from this paper to improve the design of a **mobile technology (e.g., an app, operating system, or device)**.

This can be:

- an existing technology,
- a new idea of your own, or
- a semester project you have worked on.

Use roughly 4 minutes for the presentation of the paper and roughly 2 minutes for presenting your own idea.

Important: You are expected to explicitly use relevant concepts and terminology from the course topics when presenting and discussing both the paper and your proposed design (see "list of concepts and topics").

Exam Part 2

You will engage in a 9-minute discussion of your presentation and contents of the course. Questions may refer to any concept or topic covered in the course (see “list of concepts and topics”).

You should be prepared to define concepts clearly and apply them to the paper and/or your proposed idea. Example questions include, but are not limited to:

1. **Mobile Design Challenges:** What specific challenges of designing for mobile interaction does your chosen paper and/or proposed idea address? Are there any mobile challenges that remain unaddressed? If so, how would you recommend addressing them?
2. **Design Principles:** Explain one or two interaction design principles discussed in the course. Discuss how they are (or could be) applied in the paper or your proposed idea. If they are not currently reflected in the design, explain how incorporating them might improve it.
3. **Modalities and Context:** What input and output modalities are used in the paper and/or your idea? Why are these modalities appropriate for the expected context of use? Are there other potential contexts of use where these modalities wouldn't be appropriate?
4. **Interaction Qualities:** Define the three interaction qualities discussed in the course. Which qualities are being evaluated or improved in the paper and/or your idea? Are there any important interaction qualities that are currently neglected? If so, what are the implications/consequences?
5. **Context of Use:** Describe a context of use in which the technology would work particularly well. Describe a context where it would be less suitable or problematic. Explain why.
6. **Data Types and Methods:** Classify and discuss the types of data collected in the paper's user study. What are the strengths and limitations of these choices? What alternative methods could have been used, and why?
7. **Study Design for Your Idea:** If you were to conduct a user study related to your proposed idea, what would be your study goal? What type of data would you collect, and what data collection methods would you use? What would be the procedure of the study? Explain the rationale behind these choices.

List of concepts and topics

You are expected to explicitly use relevant concepts and terminology from the course topics during your 6-minute presentation and the 9-minute discussion.

Tip: an excellent performance will show appropriate use of the course concepts and terminology across different questions, *for example*, by proactively incorporating examples and reflections about interaction qualities when answering a question about modalities or user study methods.

User study design and evaluation concepts

- Field studies vs. lab studies
- One-off vs. longitudinal studies
- Experiments
- hypotheses
- dependent and independent variables
- tasks and trials
- between/within subjects design
- Novelty effects
- Order effects
- Qualitative vs. Quantitative data
- Self-reported vs. logged/observed data
- Attitudinal vs. behavioral data
- Realism
- Limitations to the validity of the study findings (positivist worldview)
- internal validity (results answer the research questions/study goals in a trustworthy way)
- external validity (results are generalizable to other contexts)
- ecological validity (results are generalizable to real-life settings)
- Positivism vs. interpretativism (worldviews)
- User study protocol / design
- User study goal and motivation
- Participants inclusion and exclusion criteria, recruitment methods
- Procedure
- Data collection methods
- Analysis methods (statistical analysis, qualitative analysis)
- Implications for design
- Standardized questionnaires, e.g. SUS, NASA TLX
- Usability study, think aloud method

Thematic concepts:

General MOBILE HCI concepts:

- Three waves of HCI
- Challenges of designing for mobile interaction
- Interaction qualities
 - Discoverability
 - Efficiency
 - Expressiveness
- Interaction techniques
- Input/Output Modalities
 - Haptic
 - Text
 - Visual
 - Speech
 - Touch

- Etc
- Design principles
 - Reification
 - Polymorphism
 - Reuse
 - Feedforward
- Contexts of use for mobile HCI:
 - Physical
 - Temporal
 - Social
 - Task
 - Technical
- Design Heuristics (Nielsen's 10)

Mobile communication:

- selective self-presentation
- social/co-presence

Digital ecosystems:

- Walled gardens
- Artifact ecologies

Online privacy

- Privacy paradox
- Personal space
- Privacy fatigue / learned helplessness

Accessibility

- Permanent, temporary, situational impairments

Exam papers (presented in research seminars)

The following papers can be chosen to be presented at the exam.

They are also the papers that we discussed in the research seminars.

Lecture 6: Mobile Communication

- Zainab Iftikhar, Yumeng Ma, and Jeff Huang. 2023. **“Together but not together”: Evaluating Typing Indicators for Interaction-Rich Communication.** In Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 724, 1–12. <https://doi.org/10.1145/3544548.3581248>
- An, P., Zhu, J., Zhang, Z., Yin, Y., Ma, Q., Yan, C., ... & Zhao, J. (2024). **EmoWear: Exploring Emotional Teasers for Voice Message Interaction on Smartwatches.** To appear in ACM CHI 2024. <https://arxiv.org/pdf/2402.07174.pdf>

- Mariam Hassib, Daniel Buschek, Paweł W. Wozniak, and Florian Alt. 2017. **HeartChat: Heart Rate Augmented Mobile Chat to Support Empathy and Awareness.** In Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems (CHI '17). Association for Computing Machinery, New York, NY, USA, 2239–2251. <https://doi.org/10.1145/3025453.3025758>

Lecture 7: Digital Ecosystems

- Carla F. Griggio, Joanna McGrenere, and Wendy E. Mackay. 2019. **Customizations and Expression Breakdowns in Ecosystems of Communication Apps.** Proc. ACM Hum.-Comput. Interact. 3, CSCW, Article 26 (November 2019), 26 pages. <https://doi.org/10.1145/3359128>
- Carla F. Griggio, Arissa J. Sato, Wendy E. Mackay, and Koji Yatani. 2021. **Mediating Intimacy with DearBoard: a Co-Customizable Keyboard for Everyday Messaging.** In Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 342, 1–16. <https://doi.org/10.1145/3411764.3445757>
- Fanglin Chen, Kewei Xia, Karan Dhabalia, and Jason I. Hong. 2019. **MessageOnTap: A Suggestive Interface to Facilitate Messaging-related Tasks.** In Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19). Association for Computing Machinery, New York, NY, USA, Paper 575, 1–14. <https://doi.org/10.1145/3290605.3300805>
- Marta E. Cecchinato, Anna L. Cox, and Jon Bird. 2017. **Always On(line)? User Experience of Smartwatches and their Role within Multi-Device Ecologies.** In Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems (CHI '17). Association for Computing Machinery, New York, NY, USA, 3557–3568. <https://doi.org/10.1145/3025453.3025538>

Lecture 8: Privacy

- Irina Shklovski, Scott D. Mainwaring, Halla Hrund Skúladóttir, and Höskuldur Borgthorsson. 2014. **Leakiness and creepiness in app space: perceptions of privacy and mobile app use.** In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '14). Association for Computing Machinery, New York, NY, USA, 2347–2356. <https://doi.org/10.1145/2556288.2557421>
- Rebecca Balebako, Jaeyeon Jung, Wei Lu, Lorrie Faith Cranor, and Carolyn Nguyen. 2013. **"Little brothers watching you": raising awareness of data leaks**

on smartphones. In Proceedings of the Ninth Symposium on Usable Privacy and Security (SOUPS '13). Association for Computing Machinery, New York, NY, USA, Article 12, 1–11. <https://doi.org/10.1145/2501604.2501616>

- Hazim Almuhiemedi, Florian Schaub, Norman Sadeh, Idris Adjerid, Alessandro Acquisti, Joshua Gluck, Lorrie Faith Cranor, and Yuvraj Agarwal. 2015. **Your Location has been Shared 5,398 Times! A Field Study on Mobile App Privacy Nudging.** In Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems (CHI '15). Association for Computing Machinery, New York, NY, USA, 787–796. <https://doi.org/10.1145/2702123.2702210>

Lecture 9: Accessibility and Multimodal Interaction

- Carla F. Griggio, Benjamin M. Gorman, and Garreth W. Tigwell. 2024. **Party Face Congratulations! Exploring Design Ideas to Help Sighted Users with Emoji Accessibility when Messaging with Screen Reader Users.** Proc. ACM Hum.-Comput. Interact. 8, CSCW1, Article 175 (April 2024), 31 pages. <https://doi.org/10.1145/3641014> (access via: https://www.researchgate.net/publication/378070651_Party_Face_Congratulations_Exploring_Design_Ideas_to_Help_Sighted_Users_with_Emoji_Accessibility_when_Messaging_with_Screen_Reader_Users)
- Jaylin Herskovitz, Andi Xu, Rahaf Alharbi, and Anhong Guo. 2024. **ProgramAlly: Creating Custom Visual Access Programs via Multi-Modal End-User Programming.** In Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology (UIST '24). Association for Computing Machinery, New York, NY, USA, Article 85, 1–15. <https://doi.org/10.1145/3654777.3676391>
- Mayank Goel, Leah Findlater, and Jacob Wobbrock. 2012. **WalkType: using accelerometer data to accomodate situational impairments in mobile touch screen text entry.** In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '12). Association for Computing Machinery, New York, NY, USA, 2687–2696. <https://doi.org/10.1145/2207676.2208662>

Lecture 10: Quantitative Data Analysis

- Tom Biselli (Science and Technology for Peace and Security (PEASEC), Technical University of Darmstadt), Laura Utz (Science and Technology for Peace and Security (PEASEC), Technical University of Darmstadt), Christian Reuter (Science and Technology for Peace and Security (PEASEC), Technical University of Darmstadt) <https://petsymposium.org/popets/2024/popets-2024-0011.php>

Exam Preparation Assignments

The following assignments were presented during lectures 5-9 and they are meant to help you study and prepare for the presentation and Q&A of the exam. It's not necessary to hand them in.

Exam Preparation Assignment 0

1. Choose a Paper

See "[Exam papers](#)" above or go choose

from: <https://www.moodle.aau.dk/mod/page/view.php?id=2065439&forceview=1>

This will be:

- The paper you focus on in the seminars
- The paper you may later present at the exam

2. Prepare a Draft Exam Presentation (6 minutes)

Prepare a draft of your exam presentation including:

Part 1 – The Paper (~4 minutes)

- Research problem and motivation
- Key design idea (if applicable)
- User study methods
- Main findings and design implications

Part 2 – Applying the Insights (~2 minutes)

Propose your own idea for improving a mobile technology based on this paper.

This can be:

- An existing technology
- A new idea
- Your semester project

Use relevant course concepts when possible (see "[List of concepts and topics](#)")

Exam Preparation Assignment 1

Consider the design idea you proposed in your exam presentation and develop the following questions and prompts.

Remember to use relevant concepts and terminology from the course topics when answering these questions (see "[list of concepts and topics](#)").

1) Describe the intended context(s) of use of your idea. What types of context is your idea most related to?

In what ways did the paper you chose inspire/inform your answer?

2) Imagine that you want to conduct an experiment to evaluate your idea.

- What aspect of your idea would you like to evaluate?

- In what ways does the paper you chose inform/inspire this answer?
- What's your hypothesis?
- What are your independent and dependent variables?
- What task do participants need to do?
- Who would you recruit to be participants? What kind of population should they represent?
- What's the procedure of the experiment?
- How would you collect data?

3) In what ways would you control the context of use in your experiment design to make sure that it's your design what indeed causes effects on the user's performance or experience, rather than external factors?

4) In what ways would you increase realism in your experiment design, so that the results of the experiment are more likely to represent how your design would work in real life?

5) Would you call this experiment a lab experiment or a field experiment? Why?

6) Would you call this experiment a one-off study, or a longitudinal study? Why?

Exam Preparation Assignment 2

1) Consider the paper's **user study** you describe in your exam presentation and answer the following questions and prompts. Remember to use relevant concepts and terminology from the course topics when answering these questions (see ["list of concepts and topics"](#)).

- What types of data are collected in the study? Answer in terms of:
 - Behavioral vs. Attitudinal data
 - Self-reported vs. observed data
 - Qualitative vs. quantitative data
 - Conceptual question: What's the relationship between behavioral/attitudinal and self-reported/observed types of data?
- What methods are used for the data collection?
- Are there any collected data related to different types of context? If not, can you think of any data about context that could provide additional, interesting insights to the study results?

2) Consider the **design idea** you proposed in your exam presentation and develop the following questions and prompts. Remember to use relevant concepts and terminology from the course topics when answering these questions (see ["list of concepts and topics"](#)).

a) What type of study would you do to evaluate your idea or understand users' experience of it? (*Bonus challenge: if you already designed an experiment in the [Exam Preparation Assignment #1](#), propose a different type of study*)

b) Characterize the data of your study by answering the following questions:

- What types of data are collected in the study? Answer in terms of:
 - Behavioral vs. Attitudinal data
 - Self-reported vs. observed data
 - Qualitative vs. quantitative data
 - Conceptual question: What's the relationship between behavioral/attitudinal and self-reported/observed types of data?
- What methods are used for the data collection?
- Are there any collected data related to different types of context? If not, can you think of any data about context that could provide additional, interesting insights to the study results?

(Bonus challenge: if you already designed an experiment in the [Exam Preparation Assignment #1](#), answer these questions about the experiment as well)

Exam Preparation Assignment 3

Consider the paper's user study you describe in your exam presentation and answer the following questions and prompts. Remember to use relevant concepts and terminology from the course topics when answering these questions (see "list of concepts and topics").

The following paper may help with these questions: <https://www.diva-portal.org/smash/get/diva2:834169/FULLTEXT01.pdf>

1. What worldview aligns best with the study? Positivism or interpretativism (or other)? Why?
2. In what ways does the paper argue for the validity/rigor of their study?
3. If the study aligns with a positivist worldview: What are the limitations to the validity of the study findings? Consider:
 - internal validity (results answer the research questions/study goals in a trustworthy way)
 - external validity (results are generalizable to other populations and contexts)
 - ecological validity (results are generalizable to real-life settings)
4. If the study aligns with an interpretativist worldview. Consider:
 - credibility (interpretations trustworthy and well-grounded in the data)
 - transferability (findings are described richly enough that readers can judge for themselves whether they help understand other contexts)
 - reflexivity (researchers acknowledge their subjective influence in the interpretations)

Exam Preparation Assignment 4

Remember to use relevant concepts and terminology from the course topics when answering these questions (see "[list of concepts and topics](#)").

1. Consider the **design idea** you proposed in your exam presentation. In what ways it supports or lacks support for people with:
 - a. Permanent impairments?
 - b. Temporary impairments?
 - c. Situational impairments?

Recommended reading: <https://inclusive.microsoft.design/articles/inclusive-101-guidebook>

2. Consider the paper's **user study** you describe in your exam presentation.
 - a. What are the implications for design that the paper offers?
 - b. What other implications for design would you suggest, based on the study findings? For inspiration, consider:
 - i. Based on the insights from the paper, how would you improve the design of other technologies "of the same class"?
 - ii. Does the study offer relevant insights for the design of operating systems and mobile devices?
 - iii. Does any of the other papers that we discussed in class provide extra insights that, combined with this one, lead to new ideas for design?
 - iv. Are there any theories or concepts that, combined with the insights of this paper, lead to new ideas for design? (e.g., artifact ecologies or design principles)